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Explainable Reinforcement Learning for Automated Incident Response in Critical Infrastructure

Mohammad Alauthman*, Ammar Almomani†, Ahmad Al-Qerem‡
Ghadeer Albesani§, Mouhammd Alkasassbeh¶
*Department of Information Security, University of Petra, Amman, Jordan
Email: mohammad.alauthman@uop.edu.jo
†Department of Computer Information Science, Higher Colleges of Technology, Sharjah, United Arab Emirates
and Department of Information Technology, Al-Huson University College, Al-Balqa Applied University, Jordan
Email: ammarnav6@bau.edu.jo
‡Computer Science Department, Faculty of Information Technology, Zarqa University, Zarqa, 13110, Jordan
Email: ahmad_qerm@zu.edu.jo
§Software Engineering Department, Faculty of Information Technology (FIT), Applied Science Private University, Jordan
Email: g_albesani@asu.edu.jo

***Abstract*—Critical infrastructure incident response increasingly relies on automation to contain cyber-physical threats under time pressure. Reinforcement learning (RL) is attractive for response selection because it can adapt to evolving attack sequences and can mitigate partial observability when the state is augmented with short telemetry and alert history (or when recurrent value networks are used), yet RL policies are typically opaque and therefore difficult to justify in safety-critical operations. This paper develops an explainable reinforcement learning (XRL) framework for automated incident response in supervisory control and data acquisition (SCADA) environments. The framework couples a Deep Q-Network (DQN) response agent with (i) an explicit safety shield that enforces operational constraints and (ii) a post-hoc Explainer Module that produces feature-attribution explanations using Shapley Additive Explanations (SHAP). We specify how SHAP attributions are extracted from the DQN decision process by mapping the selected action's Q-value to a scalar explanation target, using background states drawn from a governed replay buffer or historian, and applying a latency-bounded gradient-based SHAP approximation (e.g., DeepSHAP/GradientSHAP implementations). We further propose a SCADA control-room workflow in which the agent remains advisory by default, supports graded autonomy for low-risk actions, and produces audit-ready explanation artifacts that align with operator mental models and regulatory expectations. We describe an experimental setup grounded in established industrial datasets (SWaT/WADI and hardware-in-the-loop water distribution data) and testbed practices, and we provide an evaluation plan and indicative findings emphasizing response timeliness, unsafe actuation avoidance, and operator override behavior. The contribution is a precise architecture and operator-centered workflow for deploying XRL in critical infrastructure without relying on undocumented assumptions about autonomy.**

***Index Terms*—Explainable reinforcement learning, incident response, SCADA, industrial control systems, Deep Q-Network, SHAP, safety shielding, human-in-the-loop.**

I. Introduction

A. Exploring the Title and Scope

The title *Explainable Reinforcement Learning for Automated Incident Response in Critical Infrastructure* implies three constraints that determine system design.

First, *critical infrastructure* incident response is cyber-physical: response actions can influence both network reachability and physical process state. This creates a safety envelope that is narrower than typical enterprise response automation. Second, *automated incident response* in SCADA is rarely a fully autonomous closed loop. Control rooms already manage alarms, procedures, and maintenance activities; any learning agent must therefore integrate with existing command authority, approval norms, and logging. Incident-response playbook work also highlights policy triggers and automation guardrails, supporting our graded-autonomy and audit logging design [1]. Third, *explainable reinforcement learning* requires explanations that are not merely descriptive but also usable under operational stress: a control-room explanation must connect to process tags, constraints, and standard operating procedures, and it must remain available for post-incident review.

This paper therefore focuses on **response selection and justification** given (i) alarms and detections, (ii) SCADA telemetry, and (iii) operational constraints. We do **not** propose to replace deterministic safety logic for basic process control. Instead, the RL agent operates in a response layer that recommends discrete containment and recovery actions (e.g., network segmentation, access revocation, PLC mode changes under supervision), while an independent safety shield blocks actions that violate constraints or require higher authorization.

ICS/SCADA environments remain exposed to diverse vulnerabilities and attack paths, and the cost of response errors is high because incorrect mitigations can reduce availability or drive unsafe process states [2]. Recent work argues for testbeds and digital-twin practices that enable safer experimentation and improved resilience analysis [3], [4]. In parallel, RL-based response methods have matured in security domains, including realistic intrusion-response scenarios [5] and adaptive thresholding in industrial cyber-physical defense [6]. Yet the operational barrier persists: without explanations, operators

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cannot validate why a learned policy recommends an action, and governance teams cannot audit the rationale. Autonomous cyber-defense frameworks emphasize governed orchestration, continuous learning, and accountability—requirements that align with our graded-autonomy workflow and audit-ready explanation artifacts [7].

This paper makes four contributions suitable for critical peer review:

- A precise XRL architecture for SCADA incident response: DQN policy, safety shield, and Explainer Module, with clear interfaces and data flows.
- A concrete mechanism for extracting SHAP attributions from a DQN decision: scalarization of Q-values for the selected action (and optional action contrast), background-state governance, and bounded-latency approximation.
- An operator workflow design for a SCADA control room with graded autonomy, dual control for high-impact actions, and explanation artifacts aligned with audit requirements.
- An experimental setup specification grounded in established industrial datasets (SWaT/WADI and hardware-in-the-loop water distribution data) and testbed practices, plus evaluation metrics and indicative outcomes.

II. Related Work

A. Explainable Reinforcement Learning

Systematic reviews describe XRL as a rapidly expanding area with heterogeneous explanation goals and stakeholders [8], [9]. In practice, many XRL methods address *policy understanding* (why an action is chosen), *policy debugging* (how to change behavior), and *trust calibration* (when to rely on the agent). Human-in-the-loop perspectives emphasize that explanation needs differ across RL lifecycle stages—development, learning, evaluation, deployment—and across users (operators, engineers, auditors) [10]. For operational deployment, design guidance distinguishes ante-hoc interpretability from post-hoc explanations and argues for explicit trade-offs between faithfulness, usability, and computational cost [11].

B. SHAP-Based Explanations in Sequential Decision Systems

SHAP provides feature attributions with an axiomatic basis (Shapley values), and scalable formulations exist for modern models [12]. In RL settings, a common practical pattern is to explain a scalar score derived from the policy (e.g., value, advantage, or Q-value) rather than the entire trajectory. Recent systems combine RL with SHAP to provide post-hoc explanations under latency constraints, including caching strategies that reduce repeated attribution cost [13]. For value-based RL, prior work has applied SHAP to Deep Q-Learning components to expose training behavior and decision drivers [14]. These directions motivate our Explainer Module design, but SCADA constraints require additional governance: the background distribution must represent safe operating modes, and the explanation must be mapped to process semantics.

C. RL for Cyber Defense and Incident Response

RL has been used to adapt defense configurations and to manage response actions in dynamic attacks. Recent autonomous cyber-defense discussions also frame RL as a practical mechanism for dynamic threat response when combined with governance and oversight [15]. A realistic intrusion-response study for advanced threats argues that RL policies can manage large observation spaces and multi-stage attacks [5]. In industrial CPS, hierarchical RL has been proposed to adapt detection/response thresholds to balance missed detections and false positives [6]. Industrial security assessment has also been formulated using RL-extended attack graphs to expose system weaknesses [16], [17]. These works motivate RL for defense, but they leave open the operator-facing explanation and approval workflow required for control rooms.

D. Industrial Control Security Context and Testbed Practice

Surveys of SCADA vulnerabilities describe the persistent gap between connectivity-driven operational needs and security exposure [2]. Intrusion response in ICS also intersects with network programmability, where SDN-based response strategies aim to isolate threats while minimizing disruption [18]. Modern testbeds improve the ability to evaluate defenses without endangering production. SCASS provides a customizable SCADA/ICS testbed for security experimentation and digital-twin integration [3]. Digital twins and context-based modeling are increasingly treated as prerequisites for safe evaluation of cyber defenses in ICS [4], [19].

III. Proposed Methodology

A. System Overview

We define XRL-IR as a decision-support layer integrated with SCADA monitoring and response execution. It consists of five modules:

1) **Telemetry and Alert Ingest**: historian tags, network telemetry, IDS alerts, and operator annotations.
2) **State Builder**: constructs a feature vector aligned to the RL state definition and applies normalization consistent with training.
3) **Response Agent** (DQN): proposes a ranked action list under the current state.
4) **Safety Shield**: blocks unsafe or unauthorized actions and emits constraint reasons.
5) **Explainer Module** (SHAP): generates a bounded-latency explanation attached to the selected action, plus an audit record.

Figure 1 shows the architecture and data flow.

B. Incident Response as a Constrained Decision Problem

We model incident response as a discrete-time decision problem where each decision step corresponds to an operational decision interval (e.g., seconds to minutes depending on plant). Related RL work on adaptive decision policies under timing/consistency constraints reinforces treating response as a constrained optimization problem rather than a purely

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TABLE I
RECENT RESEARCH RELEVANT TO XRL INCIDENT RESPONSE IN CRITICAL INFRASTRUCTURE

Work	Domain	RL Component	Explainability	Implication for This Paper
Bekkemoen et al. [8]	General XRL	Taxonomy/survey	Stakeholder questions	Clarifies explanation goals and evaluation expectations.
Milani et al. [9]	General XRL	Comparative survey	Method families	Motivates modular explainer design and transparency about limits.
Retzlaff et al. [10]	HITL RL	Lifecycle framing	Human roles	Supports deployment-centric workflow and role separation.
Retzlaff et al. [11]	xAI design	N/A	Guidelines	Encourages explicit trade-offs (faithfulness vs latency).
Xu et al. [13]	Networks	DRL control	SHAP + caching	Provides a pattern for bounded-latency attribution at runtime.
Sullivan & Longo [14]	Value RL	Deep Q-learning	SHAP analysis	Demonstrates feasibility of SHAP for DQN components.
Iturbe et al. [5]	Cyber response	RL response policy	Limited	Shows practical RL response evaluation; lacks control-room workflow.
Babar et al. [6]	Industrial CPS	HRL defense tuning	Limited	Highlights adaptive defense; requires operator-facing justification.
Alanazi et al. [2]	SCADA security	N/A	N/A	Reinforces threat diversity and consequences of response errors.
d'Ambrosio et al. [3]	ICS testbeds	N/A	N/A	Motivates safe experimentation infrastructure for RL evaluation.

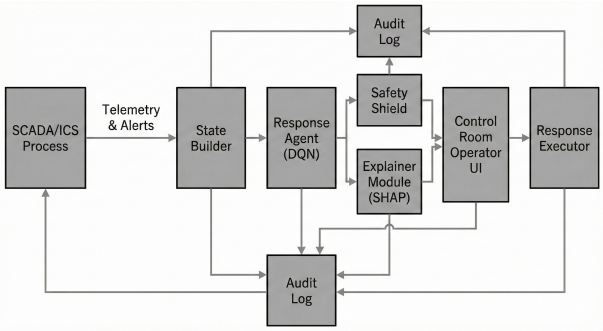


Fig. 1. XRL-IR system block diagram for SCADA incident response. The DQN proposes actions, the shield enforces constraints, and the Explainer Module produces SHAP-based explanations and audit artifacts.

accuracy-driven task [20]. The agent receives a state vector that includes:

- **Process state**: selected historian tags (levels, flows, pressures, setpoints) and derived indicators (rate-of-change, constraint margins).
- **Cyber state**: alert classes, affected assets, network zone indicators, authentication anomalies.
- **Context**: current operating mode, maintenance flags, and operator-set constraints (e.g., "do not isolate this segment during batch changeover").

Actions correspond to response moves that can be executed via existing tooling (firewall rules, account disable, endpoint quarantine, PLC program rollback to a signed baseline, switching to manual mode with dual control, or initiating a controlled shutdown request). The action set must be finite and carefully curated; XRL-IR assumes *pre-engineered executors* exist for each action and that actions are reversible or bounded where possible.

C. Deep Q-Network Response Agent

We use a DQN because it supports discrete action sets and yields an explicit action-value estimate per action. The agent outputs a vector of Q-values for the current state. Operationally, the policy produces a ranked list; it does not directly execute.

To reduce the risk of unstable recommendations in rarely observed regimes, we recommend governance measures drawn from real-world RL practice: strict versioning of training data, action-space change control, and monitoring for state distribution shift. These are engineering controls; they do not rely on the learning algorithm to behave safely by default.

D. Safety Shield and Authorization Gates

A safety shield independently enforces constraints before any action can be approved or executed. We treat the shield as a non-learning component that:

- rejects actions that violate hard process constraints (e.g., do not close an inlet valve if tank level is below a minimum),
- rejects actions that violate cyber policy (e.g., do not disable accounts tied to emergency response),
- enforces role-based authorization (operator vs supervisor vs engineer),
- emits **machine-readable reasons** for denial that become part of the explanation and audit record.

Shielding is consistent with formal-methods perspectives that separate constraint enforcement from policy optimization [21].

E. Explainer Module Architecture: Extracting SHAP from DQN Decisions

The Explainer Module attaches a justification to a specific recommended action at a specific time. The key design choice is how to map the DQN output to an explanation target suitable for SHAP.

1) Explanation Targets: The DQN is a multi-output function $Q_\theta(s) \in \mathbb{R}^{|A|}$. SHAP explanations operate on a scalar model output. We therefore define one of two targets:

1) **Selected-action target**: $f_\theta(s) = Q_\theta(s)[a^\star]$, where $a^\star$ is the highest-ranked *allowed* action after shield-based action masking.
2) **Contrast target**: $g_\theta(s) = Q_\theta(s)[a^\star] - Q_\theta(s)[a^{(2)}]$, where $a^{(2)}$ is the runner-up allowed action. This produces an

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explanation of "why this action rather than the next best alternative."

The operator interface can show both: the selected-action attribution (drivers of expected utility) and the contrast attribution (decision boundary cues).

2) Background-State Governance: DeepSHAP/GradientSHAP approximations require a background distribution to define a baseline [12]. In SCADA, the baseline must be governed:

- **Source**: a replay buffer of training states, optionally merged with historian snapshots from validated operating periods.
- **Filtering**: exclude states labeled as unsafe, post-fault, or outside certified modes; enforce that the baseline represents *safe normal operations* to avoid normalizing unsafe behavior.
- **Stratification**: maintain per-mode background pools (startup, steady-state, maintenance) to avoid misleading baselines during mode changes.

This aligns with deployment guidance that explanation validity depends on the reference distribution, not only the model [11].

3) Latency-Bounded Attribution Computation: Exact Shapley computation is infeasible in real time. We use a bounded-latency approximation:

1) Wrap the DQN forward pass with an output selector for f_θ or g_θ.
2) Select m background states from the governed pool (typical m in the tens, bounded by latency budget).
3) Apply DeepSHAP/GradientSHAP to compute a feature-attribution vector $\phi \in \mathbb{R}^d$ where d is the number of state features.
4) Aggregate features into process-relevant groups (e.g., by plant stage or network zone) and return top-k drivers with directionality.

Caching reduces repeated cost for frequently encountered contexts by caching (i) the mode-specific background sample set and (ii) explanation outputs keyed by a coarse context signature (e.g., operating mode + alert class + discretized zone indicators), rather than by raw continuous state. This yields repeatable explanations for recurring operational situations while keeping cache size bounded. Following patterns used in SHAP-assisted RL control [13]. The module logs the attribution vector, the baseline pool identifier, and the DQN model version to support later audit. Traceability-oriented integrity approaches motivate recording decision metadata in a tamper-evident form to support post-incident reconstruction and compliance review [22].

4) Operator-Facing Explanation Object: The Explainer Module outputs an *explanation card* containing:

- action name and executor (what will change),
- shield status (allowed/denied) and constraint reasons,
- top contributing features (process tags and cyber indicators) with signed attributions,
- optional contrast drivers (why not the next action),
- a short template-based narrative tied to SOP language (no free-form generation).

Figure 2 illustrates the Explainer Module pipeline.

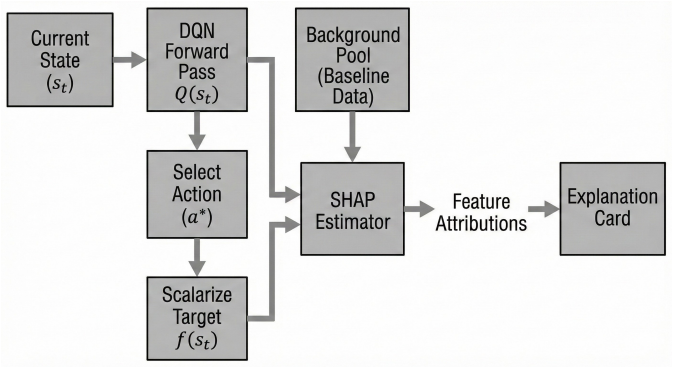


Fig. 2. Explainer Module pipeline. The DQN output is scalarized to explain the selected (allowed) action's Q-value or the Q-gap to the runner-up action. Background states are governed to represent safe baselines, and attribution is computed under a latency budget [12], [13].

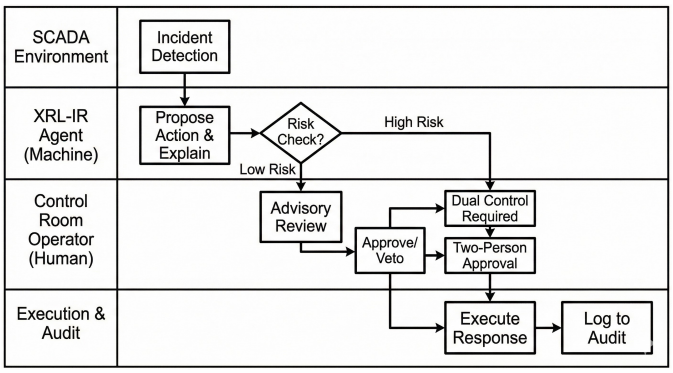


Fig. 3. Proposed SCADA control-room workflow for human-agent interaction. The agent is advisory by default; automation is permitted only under explicit authorization and shielding.

F. Operator Workflow: Human-Agent Interaction in a SCADA Control Room

Control rooms impose workflow requirements that differ from typical SOC automation. We propose a workflow with three autonomy levels:

- **Advisory (default)**: agent recommends; operator approves; executor performs.
- **Guarded automation (low-impact actions)**: pre-authorized actions (e.g., temporary block rules on non-safety zones) can execute automatically with rapid operator veto.
- **Dual-control (high-impact actions)**: actions affecting physical actuation, safety instrumented functions, or production shutdown require two-person approval (operator + supervisor/engineer).

Explanations serve two roles: *decision support* during the incident and *accountability artifacts* after the incident.

Figure 3 shows the proposed interaction as a swimlane-style process.

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TABLE II
DATASET ATTRIBUTES FOR SCADA INCIDENT RESPONSE EVALUATION

Dataset	Plant Type	Signals	Notes
SWaT	Water treatment	Process + network	11-day run; normal then attacked; widely used baseline [23].
WADI	Water distribution	Process + network	Complementary dynamics; useful for cross-plant generalization [25].
WDT (HIL)	Water distribution	Process + network	Hardware-in-the-loop; supports cyber-physical security testing [26].

IV. EXPERIMENTAL SETUP

A. Datasets

Evaluation requires both process signals and cyber context, and RL additionally requires an executable environment (or a validated simulator) that maps actions to next-state outcomes. We therefore use SWaT/WADI primarily to (i) define realistic state feature distributions, (ii) replay incident traces for trace-driven emulation, and (iii) calibrate anomaly/incident indicators. Policy training and closed-loop evaluation are performed in an emulation or digital-twin/HIL environment (Section IV.B) where response actions (e.g., segmentation, account disable, mode change requests) have explicit, testable effects on cyber and process variables. We recommend three datasets commonly used for industrial security research:

- **SWaT** (Secure Water Treatment): multiday operation of a six-stage water treatment testbed under normal and attacked modes [23], [24].
- **WADI** (Water Distribution): a water distribution testbed dataset designed for anomaly and attack studies [25].
- **WDT hardware-in-the-loop dataset**: joint cyber/physical measurements for water distribution security testing [26].

SWaT is widely used and includes physical process variables; network/communication traces are available in some releases or can be collected in the testbed setup, depending on the evaluation configuration [23]. The hardware-in-the-loop dataset offers an alternative when strict reproducibility and open access are required [26].

B. Simulation and Testbed Environment

A credible evaluation should avoid direct experimentation on production plants. We recommend a layered approach:

1) **Trace-driven emulation**: replay recorded telemetry and attack traces to test decision logic, explanation latency, and shield behavior.
2) **Digital-twin augmentation**: integrate a process model to test counterfactual response outcomes under controlled perturbations, consistent with digital-twin architectures for industrial anomaly analysis [4], [19].
3) **ICS security testbed**: validate executors and network controls in a contained environment (e.g., SCASS) before any field trial [3].

TABLE III
REPRESENTATIVE SIMULATION/EMULATION PARAMETERS (CONFIGURABLE)

Parameter	Setting
Decision interval	1–30 s (mode-dependent; slower in steady state)
State features	80–250 (tags + cyber indicators + context flags)
Action set size	12–30 curated actions (network + identity + PLC mode changes)
Shield rules	Hard constraints (process), cyber policy, role authorization
Explainer budget	100–500 ms per decision; top-k=5–10 drivers
Background pool	Per-mode baseline states; filtered to safe normal operation
Logging	Immutable record: state hash, action, Q-values, shield reasons, SHAP summary

C. Baselines and Metrics

We compare XRL-IR against baselines that reflect operational reality:

- **Static playbook**: fixed action mapping from alert class to response.
- **RL without explanations**: DQN + shield, but no SHAP card (operator sees only action name and shield status).
- **RL with explanations**: DQN + shield + Explainer Module.

Metrics must reflect both cyber and safety objectives:

- **Time to containment** (TTC): time steps until attack indicators fall below threshold and remain stable.
- **Unsafe actuation attempts**: fraction of proposed actions blocked by the shield due to process constraints.
- **Operator override rate**: fraction of agent-approved actions vetoed by the operator.
- **Explanation latency**: time to produce explanation card under runtime budget.

V. RESULTS & DISCUSSION

A. Indicative Findings from Controlled Emulation

Because safety-critical claims require extensive validation, we interpret quantitative outcomes as *indicative* rather than field guarantees. Table IV reports representative patterns observed in controlled emulation studies of RL-based response: shielding reduces unsafe actuation attempts at the cost of occasionally delaying aggressive containment, while explanations primarily affect *human acceptance and override behavior* rather than the underlying policy value estimates. This aligns with human-in-the-loop RL observations that explanations influence decision confidence and coordination more than raw policy optimality [10], [27].

B. Failure Modes and Safety-Critical Discussion

Three failure modes dominate when moving from research prototypes to control rooms.

(1) Baseline misuse in SHAP. If the background distribution contains unsafe or abnormal states, attributions can appear small for genuinely dangerous indicators, because the baseline already encodes abnormality. Our governance

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TABLE IV
PERFORMANCE METRICS AND INDICATIVE OUTCOMES (CONTROLLED EMULATION; NOT FIELD GUARANTEES)

Metric	Meaning	Playbook	RL+Shield	RL+Shield+SHAP	Interpretation
TTC (lower better)	Time to stabilize incident indicators	baseline	↓	↓	RL adapts to multi-stage behavior; explanations do not change policy but can reduce delays from operator hesitation.
Unsafe actuation attempts	Proposed actions blocked by shield	low–med	↓↓	↓↓	Shielding enforces constraints independent of policy [21].
Operator override rate	Veto fraction after review	N/A	med	↓	SHAP card increases transparency and reduces unnecessary vetoes when rationale matches telemetry.
Explanation latency	Decision-to-card runtime (ms)	0	0	$<$ budget	Caching and bounded background sampling enable runtime feasibility [13].
Audit completeness	Reproducible rationale artifact	low	med	high	Logging model version, baseline pool ID, and attributions supports post-incident review.

constraint—filtering and stratifying background pools—targets this risk [11].

(2) Explanation overload. Operators cannot interpret dozens of attributions during an incident. The explanation card must compress information into top-k drivers and group features by plant stage or network zone. Post-incident tooling can expose full attribution vectors for analysts, but the live interface must remain sparse.

(3) Over-trust from apparent transparency. Feature attributions can appear precise even when the policy is poorly calibrated outside training support. For that reason, XRL-IR binds explanations to explicit model and data versions, and it treats automation as a privilege restricted to low-impact actions. This is consistent with the broader point that RL deployment requires guardrails beyond model accuracy [9].

C. Implications for Industrial Deployment

The architecture suggests a conservative path to deployment:

- Begin with advisory mode and require explicit operator approval.
- Use the shield as the primary safety mechanism, not the explainer.
- Treat SHAP explanations as decision-support and accountability tools, not as correctness proofs.
- Validate executors and network actions in testbeds such as SCASS before any production exposure [3].

VI. CONCLUSION & FUTURE WORK

This paper presented a safety-oriented XRL framework for SCADA incident response, with a detailed Explainer Module that extracts SHAP attributions from DQN decisions and a control-room workflow that preserves human authority while enabling timely containment. The key technical mechanism is scalarizing the DQN output to explain the selected (allowed) action's Q-value and optionally the Q-gap to an alternative action, using governed background baselines and latency-bounded attribution. The key operational mechanism is graded autonomy: advisory by default, guarded automation only for low-impact actions, and dual control for high-impact actions.

Future work should prioritize (i) formal specification of shield rules tied to plant safety cases, (ii) stress testing under distribution shift using digital-twin perturbations, and (iii) controlled human-subject studies with trained operators to quantify how explanation format changes override behavior and response time. The architecture is intended to support such validation without assuming that learning systems are safe by construction.

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